Year 12 Maths - Applications of differentiation

Similar questions - Working with gradient functions

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A curve C satisfies $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 6x - 12$. At the point where $x = 0$, the gradient of C is 9, and C passes through $(2, 5)$.

(a) Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ and then find the equation of C. **[5]**

(b) Find the coordinates of the stationary points of C and determine their nature. **[4]**

(c) Show that C has a point of inflection at $x = 2$, and find its coordinates. **[3]**

2 The curve $y = px^2 + qx$, where p and q are constants, has a minimum point at $(3, -18)$.

(a) Use calculus to show that $q = -6p$, and find the values of p and q. **[5]**

(b) Find the area of the region enclosed by the curve and the x-axis. **[4]**

Answers

Question 1

(a) $\dfrac{dy}{dx} = 3x^2 - 12x + c$; at $x = 0$ the gradient is 9, so $c = 9$. $y = x^3 - 6x^2 + 9x + d$; through $(2,5)$: $8 - 24 + 18 + d = 5$, so $d = 3$. $y = x^3 - 6x^2 + 9x + 3$

(b) $3(x-1)(x-3) = 0$: $x = 1$ gives $(1,7)$, and $x = 3$ gives $(3,3)$. $\dfrac{d^2y}{dx^2} = 6x - 12$ is $-6 < 0$ at $x = 1$ (maximum) and $6 > 0$ at $x = 3$ (minimum)

(c) $\dfrac{d^2y}{dx^2} = 0$ at $x = 2$ and changes sign from negative (e.g. $x = 1.5$) to positive ($x = 2.5$). Point $(2,5)$

Question 2

(a) $\dfrac{dy}{dx} = 2px + q = 0$ at $x = 3$, so $6p + q = 0$ and $q = -6p$. Also $9p + 3q = -18$, so $9p - 18p = -18$, giving $p = 2$, $q = -12$

(b) $2x^2 - 12x = 0$ at $x = 0, 6$. $\displaystyle\int_0^6 (2x^2 - 12x)\,dx = \left[\tfrac{2}{3}x^3 - 6x^2\right]_0^6 = 144 - 216 = -72$. Area $= 72$